UNIVERSITY OF UYO

DEPARTMENT OF MATHEMATICS

FIRST SEMESTER EXAMINATION

MTH 211: MATHEMATICAL METHODS I

2025/2026 SESSION

INSTRUCTION: Attempt all the questions

TIME: 1hr: 40 minutes

1. Using the Mean Value Theorem (MVT) find the value(s) of a point, c, at which the function $f(x) = 3x^3 - x^2 + 5x$ obeys the MVT, if $x \in [-1,3]$.

2. Prove that the $\lim_{(x,y,z)\to(1,2,-1)} (3x^2 + 2y^2 - z) = 12$.

3. Obtain by means of differentials the approximate value of $\sqrt{38}\sqrt{732}$.

4. Find the Taylor's series expansion of the function, $f(x) = \text{Cos}xy$ up to three terms about the point
$(x_0, y_0) = (\frac{\pi}{3}, 1)$

5. If $x = r\cos\theta$, $y = r\sin\theta$, show that $\frac{\partial(x,y)}{\partial(r,\theta)} \cdot \frac{\partial(r,\theta)}{\partial(x,y)} = 1$

6. (a). If the trajectory of a particle in space has a parametric form $x = e^t$, $y = 2\sin 3t$ and $z = 4\cos 3t$, find the magnitude of its velocity and acceleration vector at $t = 0$
(b). Find the directional derivative of the function $f(x, y) = x^2y^2 - 4y$ at the point $(2, -1)$ in the direction of the vector $v = 2i + 5j$.

7. Find the divergence and curl of the vector field $\vec{v} = (x^2 - y^2)i + 2xyj + (y^2 - xy)k$.